The complexity arrow tracks spectral rigidity — as a trend, not a law:
a pre-registered falsifier test on Krylov chains, random-matrix ensembles, and the DSSYK chord basis

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29 July 2026

Abstract

Krylov (spread) complexity grows on a semi-infinite chain whose position operator is positive with a boundary at zero — a structure that in double-scaled SYK (DSSYK) is not an analogy but an identity: the thermofield double's Krylov basis *is* the chord-number basis, so complexity equals the expected chord number [1]. We ask what property of a Hamiltonian's *spectrum* controls this complexity arrow, and we ask it as a pre-registered falsifier test. Across seven ensemble families — GOE, Poisson, Anderson chains, β-Hermite tridiagonal, Rosenzweig–Porter, banded, and the deterministic DSSYK chord chain — with spectral rigidity measured from independently unfolded finite spectra (never from the Jacobi data driving the complexity), we find: (1) no counterexample to rigidity–arrow co-variation, including the ensemble designed to break it; (2) a forced refinement — the DSSYK chord chain is *picket-fence* rigid (gap ratio $r = 0.968$, integrable-type, not random-matrix) yet carries the strongest arrow, so the arrow tracks *rigidity*, not chaos; chaos is one route to rigidity; (3) quantitatively, the trend is real (Spearman $\rho = -0.765$, $p = 3.4 \times 10^{-5}$, pre-registered) but *universality fails exactly as its pre-registered falsifier anticipated*: at matched rigidity, ensemble families branch by an order of magnitude in complexity, and the branch variable is eigenvector localization (inverse participation ratio). The honest conclusion: *spectral rigidity predicts the complexity arrow as a trend but is not sufficient; eigenvector delocalization completes the relation.* Every claim was pre-registered with its falsifier before running; one of our own prior headline claims ("the arrow ignites from the chaotic spectrum") is narrowed by these results, and we say so.

1 Setup: the arrow lives on a one-sided chain

The Lanczos recursion maps any Hamiltonian and seed state to a semi-infinite tridiagonal chain; spread complexity $K(t) = \langle \hat{n} \rangle$ is the wavepacket's mean position, a manifestly positive operator with a boundary at $n = 0$. In DSSYK this chain is physical: the chord basis [2, 3] makes the Hamiltonian tridiagonal with hopping $b_n = \sqrt{(1-q^n)/(1-q)}$, and the thermofield double's Krylov basis coincides with the chord-number basis exactly, at any q [1] — complexity *is* chord number, and (in the triple-scaling limit) bulk wormhole length; at finite N the chord basis is an extrapolation and complexity saturates [4]. Our exact chain lab confirms the registered expectations: $K(t) = b_1^2 t^2$ at small t (to $< 2 \times 10^{-5}$); ballistic pre-boundary transport at every q with velocity set by the hopping plateau $b_\infty = 1/\sqrt{1-q}$; a super-ballistic ($K \sim t^2$) contrast in the $q \to 1$ oscillator limit; all readouts strictly inside the pre-reflection window, so no finite-cutoff artifact is claimed as physics.

2 The falsifier design

The Lanczos coefficients determine both the spectral density and $K(t)$, so "complexity relates to the spectrum" is trivially true. To make the question falsifiable we required: (i) spectral rigidity measured from *independently unfolded finite spectra* — gap ratios r and number variance $\Sigma^2(L)$ — never reconstructed from the same Jacobi data that evolves $K(t)$; (ii) ensembles chosen to break the hypothesis if it can be broken, in particular β-Hermite tridiagonal matrices (disordered Lanczos coefficients with exactly random-matrix spectra) and Rosenzweig–Porter interpolation through its transition; (iii) verdicts pre-registered before each run, with the falsifier stated. An independent analytical reviewer certified the design non-circular before any code ran.

3 Stage one: no counterexample, one refinement

Across GOE, Poisson, Anderson, β-Hermite, Rosenzweig–Porter, banded, and DSSYK-chord ensembles (two independent executions; dimensions 64–128, up to 64 disorder samples), no clean bridge-killer appeared: rigid spectra came with delocalized chains and strong late complexity, Poisson-like spectra with localized chains and weak complexity, intermediate ensembles intermediate on both axes. The designed killer survived: β-Hermite's coefficient disorder is relatively weak and both its columns stay random-matrix-like.

The refinement came from our own arena. The DSSYK chord chain's spectrum has $r = 0.968$ with low number variance: near-equally-spaced levels — *picket-fence* rigidity, characteristic of integrable spectra like the oscillator, categorically distinct from random-matrix level repulsion ($r \approx 0.53$–0.60) — yet it carries the strongest arrow in the comparison. The classifier was patched to separate picket-fence from random-matrix rigidity, and the bridge statement was narrowed accordingly:

> *The complexity arrow co-varies with spectral rigidity (low number variance), not with chaos per se. Chaos is one route to rigidity; the deterministic chord chain is another.*

This narrows our own earlier claim ("the arrow ignites from the chaotic spectrum"): chaotic should have been *rigid*. The chord chain, being a single deterministic instance, is kept in all comparisons as a labeled $n = 1$ structured control, not as an ensemble member.

4 Stage two: the trend is real; universality fails, informatively

We then pre-registered a quantitative test over 23 configurations spanning the seven families (parameter sweeps in disorder strength, β, coupling, bandwidth, and q), with all measurement code identical to stage one:

Q-1 (trend) — confirmed. Normalized late complexity K_{late}/chain versus $\Sigma^2(L{=}4)$ is monotone-decreasing across the ensemble scatter: Spearman $\rho = -0.765$, $p = 3.4 \times 10^{-5}$ at dimension 128 with 32 samples per point (independent smoke run at lower power: $\rho = -0.743$).

Q-2 (universality) — refuted, exactly as pre-registered. If rigidity *sufficed*, points from different families at matched Σ^2 would agree. They do not, in any populated rigidity bin: e.g. at $\Sigma^2 \approx 3.5$–4.0, an Anderson chain ($W{=}5$) gives K_{late}/chain $= 0.032$ while Rosenzweig–Porter ($c{=}0.5$) gives 0.481 — a factor ~ 15 at matched rigidity, far beyond combined uncertainties. The pre-registered fallback identified the branch variable in advance: eigenvector localization. The recorded inverse participation ratios at that matched point are 0.328 (localized) versus 0.067 (delocalized).

Q-3 (control) — consistent. The deterministic chord points sit at the rigid end of the scatter with strong complexity ($q = 0.9$ is the most rigid configuration in the entire dataset, $\Sigma^2 = 0.40$).

5 Conclusion, stated honestly

> *Spectral rigidity predicts the complexity arrow as a trend ($\rho \approx -0.77$) but is not sufficient: there is no universal $K_{\mathrm{late}}(\Sigma^2)$ curve. Eigenvector delocalization is the additional variable that completes the relation.*

Three scope statements. First, these are finite-dimensional numerical results (shadows of the type-III_1/large-N statements they gesture at); a falsifier hunt that finds no counterexample is support, not proof. Second, the DSSYK chord chain enters as a deterministic labeled control; its picket-fence rigidity is an observation about one physically central instance, not an ensemble statement. Third, the underlying conjecture that motivated this study — that the one-sided chord chain's positivity ($\hat{n} \geq 0$ with a boundary) is the finite shadow of a half-sided modular/arrow-of-time structure in DSSYK — remains exactly that, a conjecture; our literature verification found the positivity-as-arrow fusion unoccupied (positivity is used thermodynamically in [6], modular structure algebraically in [7]), and nothing here proves it.

Methods — pre-registration and three-agent verification

Every stage followed the same discipline: predictions and falsifiers registered in the public repository before running; measurement code frozen across stages (the quantitative stage calls the stage-one measurement function verbatim, sweeping only parameters); spectra unfolded independently of the chain data; two independent executions (author and clean-room, the latter from the formulas alone) for every load-bearing number; and an analytical reviewer's audit at each gate (non-circularity of the design; the b_n symmetrization; the picket-fence reading; the final conclusion's wording). Two claims of ours died in the process — the universality of $K_{\mathrm{late}}(\Sigma^2)$, and the "chaotic spectrum" phrasing of our own earlier capstone — and both deaths are reported as results. Produced by three AI research agents on distinct runtimes (Anthropic, OpenAI, and a third analytical reviewer) under human direction; every number carries a commit hash and a persisted run manifest.

References

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Acknowledgements. Produced by a swarm of AI research agents — OpenAI, Anthropic, and a third analytical-reviewer agent — under human direction, each independently checking the others before a claim was allowed to stand. Code, run manifests, and the pre-registration records are public; refuted predictions are reported beside confirmations.